



UNIVERSITI TUNKU ABDUL RAHMAN

Tutorial 3

1. Explain the consequences of a project when the manager allows the scope of a project to gradually increase.
2. A project has been defined to contain the following list of activities along with their required times for completion. You are required to

Activity	Immediate Activity	Weeks	Predecessors
1	Collect requirements	2	--
2	Analyze processes	3	1
3	Analyze data	3	2
4	Design processes	7	2
5	Design data	6	2
6	Design screens	1	3, 4
7	Design reports	5	4, 5
8	Program	4	6, 7
9	Test and document	8	7
10	Installation	2	8, 9

- (a) Draw a network diagram for the activities.
- (b) Fill in the activity summary table below

Activity	Time	Immediate Predecessors	Early Start	Early Finish	Late Start	Late Finish	Total Slack	Critical Path?
1	2	--						
2	3	1						
3	3	2						
4	7	2						
5	6	2						
6	1	3, 4						
7	5	4, 5						
8	4	6, 7						
9	8	7						
10	2	8, 9						

- (c) Calculate the earliest expected completion time.
- (d) Show the critical path.

3. Assume that you have a project with seven activities labeled A-G (below). Derive the early start, (ES) early finish (EF), late start (LS), late finish (LF) time of the project, total slack time for each of the following tasks (begin at time =0). Which tasks are on the critical path?

Activity	Preceding Event	Expected Duration	ES	EF	LS	LF	Total Slack	Critical Path?
A	-	5						
B	A	3						
C	A	4						
D	C	6						
E	B, C	4						
F	D	1						
G	D, E, F	5						